

Designing for Justice

COLLEGE 117
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Recap of Monday

Ethics of Research

- Emerged in reaction to historical events
- Harms can still occur even with ethical oversight mechanisms, regulations, and rules
    - Limited attention to downstream implications, especially social harms (e.g., AI)
- Research leads to development of biotechnology
- Often public & downstream communities are not included as decision-makers in what makes research ethical

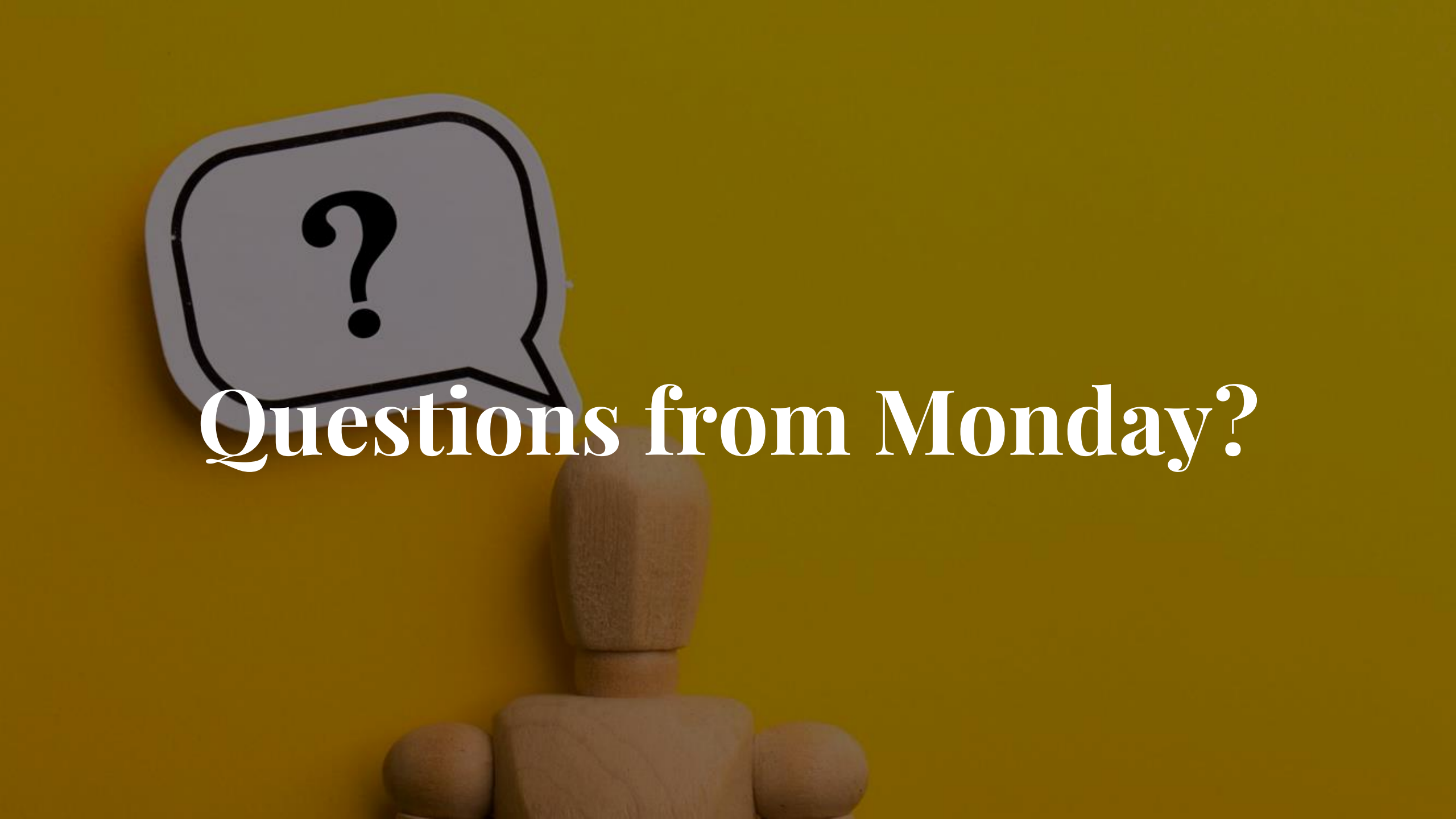


Recap of Monday

Engagement

- Part of the ‘fair, deliberative process’ Daniels calls for to meet health needs
- A way to help researchers & tech developers identify who receives the benefits and who bears the burdens
- Arguments for it include:
  - Giving a voice
  - Identifying risks & benefits
  - Honoring the ‘right’ to engage (i.e., social justice)
  - Utilizing a more ethical and trustworthy approach




Questions from Monday?



Today's guiding questions

1. How can technology be designed for justice?

2. "Who and what are fixed in place–classified, corralled, and/or coerced–to enable technoscientific development? " (Benjamin, 2016)

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Technology does not exist in and of itself. It is a tool, or range of tools for doing something

Technological innovation and justice are sometimes seen as conflicting priorities that cannot be met simultaneously.

Partner discussion:

"What people came to realize is that before we can have affordable computers, we must have computers. The greater good is the benefit of this, not the small individual tinkering, which will slow the whole thing down." (Benjamin, 2013, pp. 162)

Is this the kind of inequality Rawls would tolerate?

Designing for justice

Decision-Making

- Upstream design decisions impact downstream outcomes

When in the technology development process does a given actor have opportunities to participate in decisions? (Harsh, 2011)

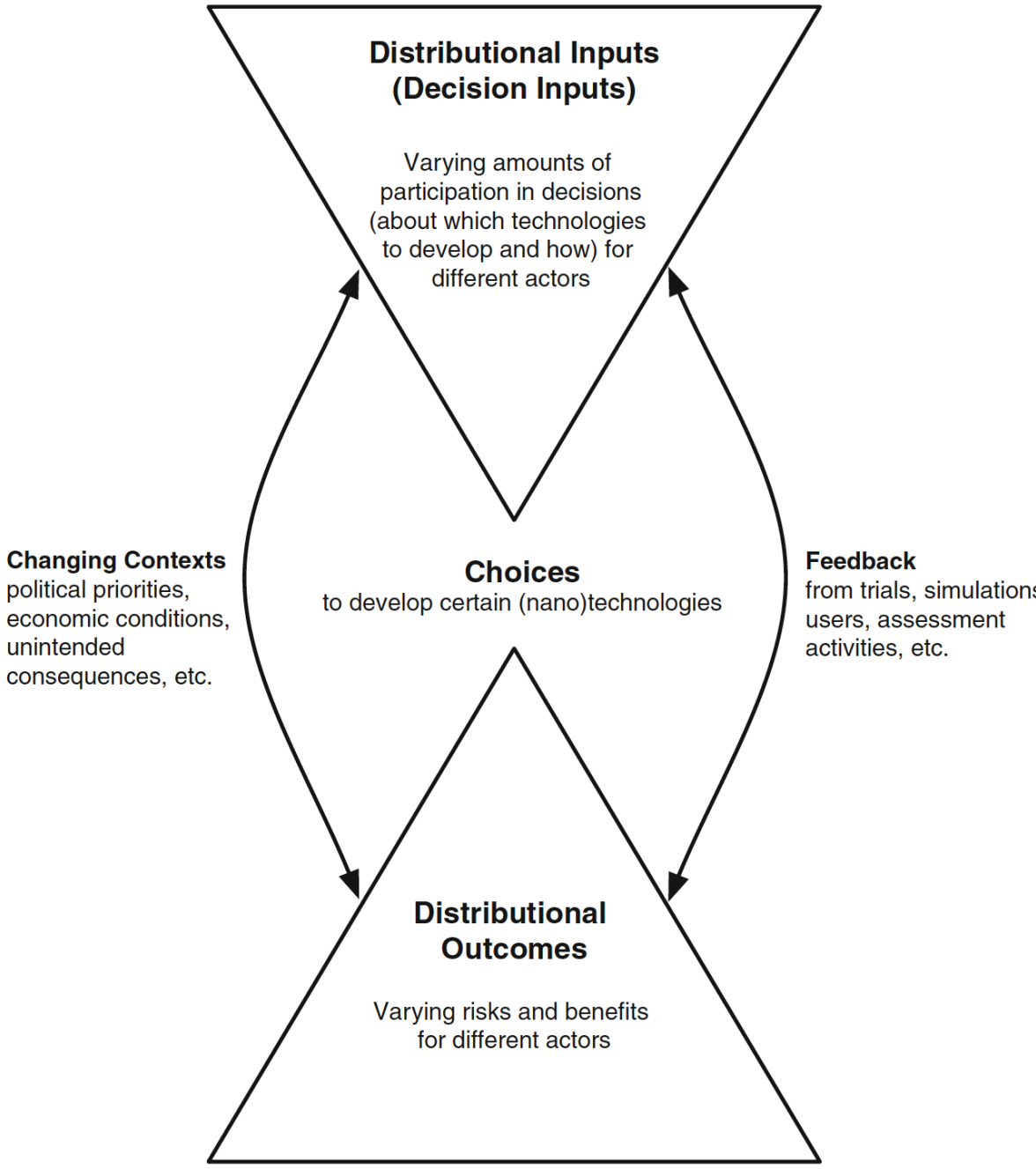


Fig. 15.1 A distributional view of technology development

Designing for justice

Time & Context

- Baked into tech development

"No matter what nanotechnologies are being developed, or in which developing region they are being developed, they will be developed among existing inequalities that will affect access to decisions." (Harsh, 2011)

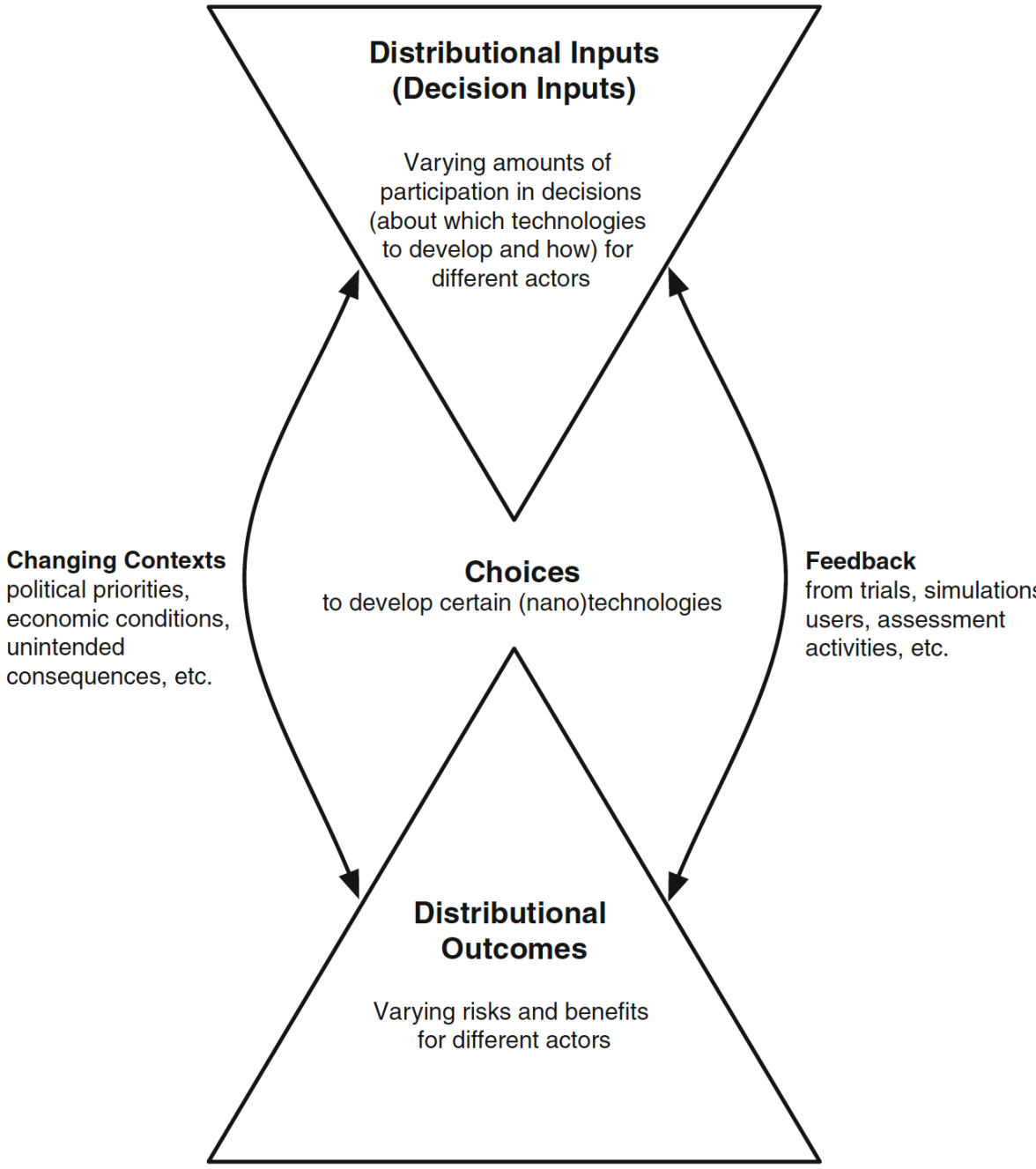


Fig. 15.1 A distributional view of technology development

Why might engagement be important when designing for justice?

Nobody has responded yet.

Hang tight! Responses are coming in.

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Distributive Justice

The equitable, or fair, allocation of resources, opportunities (including decision-making opportunities), and benefits within a society or context

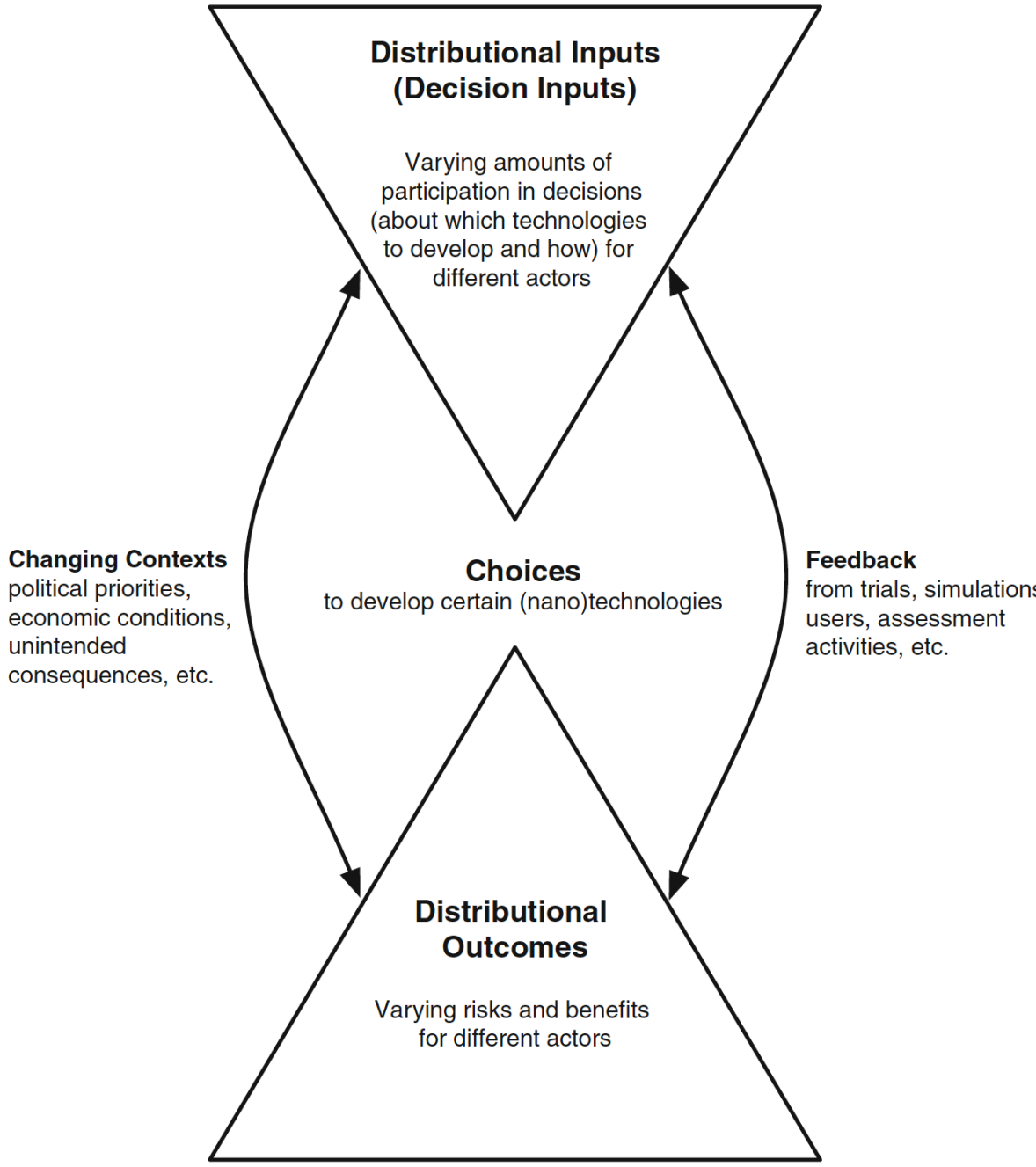


Fig. 15.1 A distributional view of technology development

Participatory Justice

The direct participation of those affected by a particular decision in the decision-making process.

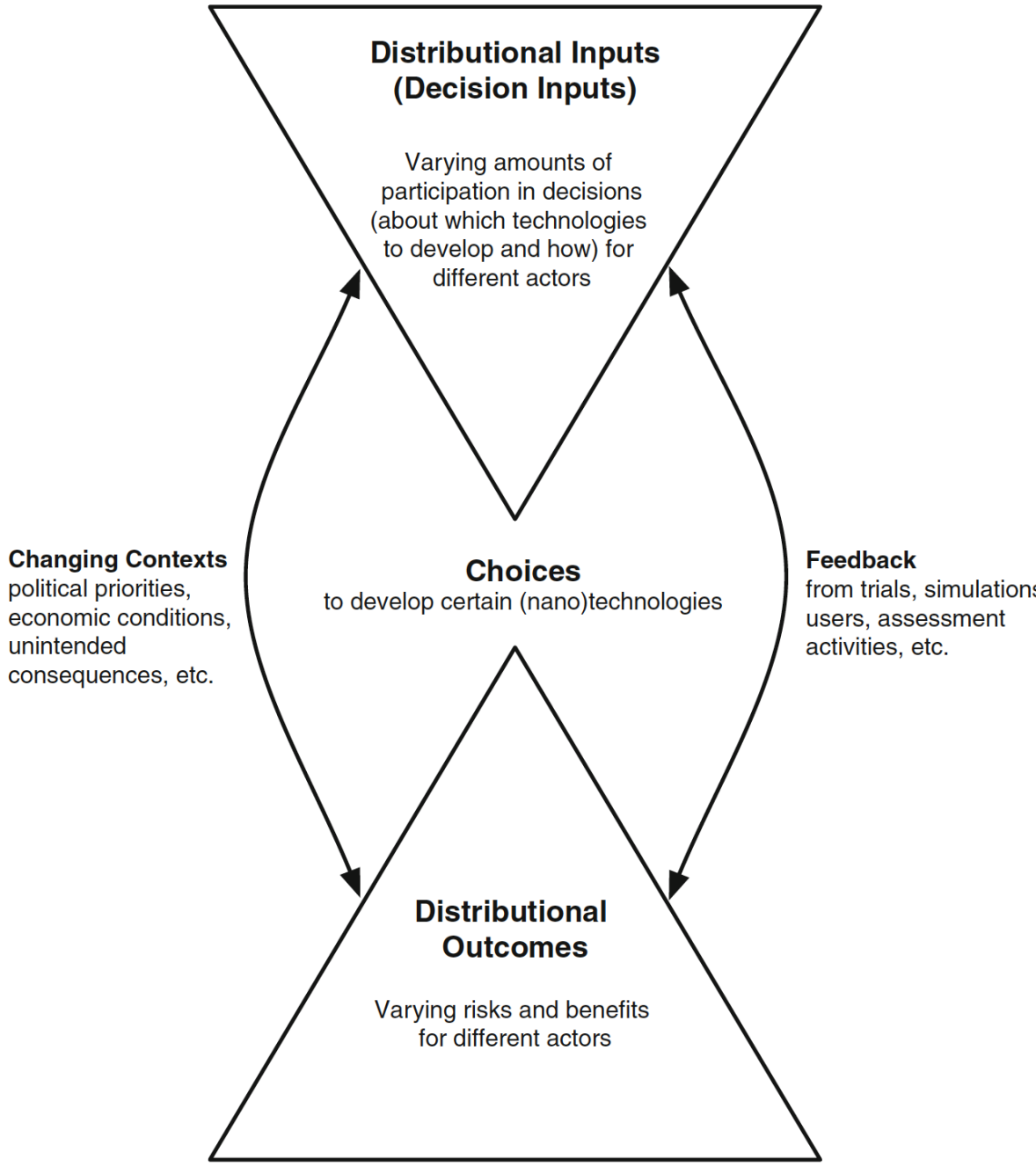


Fig. 15.1 A distributional view of technology development

Inclusive design: designing for individuals who are traditionally excluded to build **frameworks that work better for everyone**



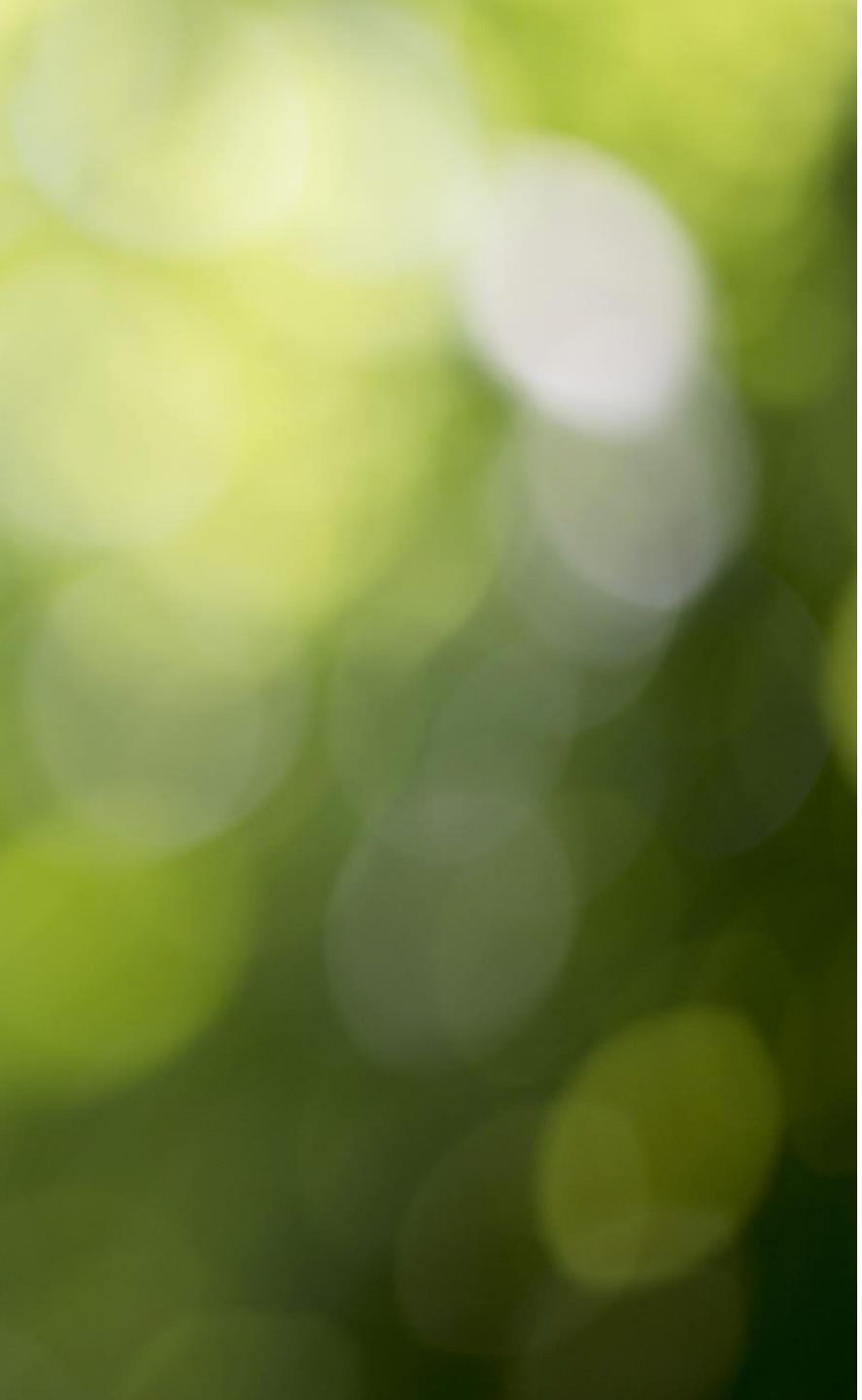
Inclusive Design

- A human-centered design process
- Embraces difference and leverages it to shape decisions
- The antithesis of the pull to ‘simplify’
- Requires awareness of the context and broader impact of decision-making



https://legacy.idrc.ocadu.ca/about-the-idrc/49-resources/online-resources/articles-and-papers/443-whatisinclusivedesign


Design
Council
Inclusive Design
from the pixel to the city



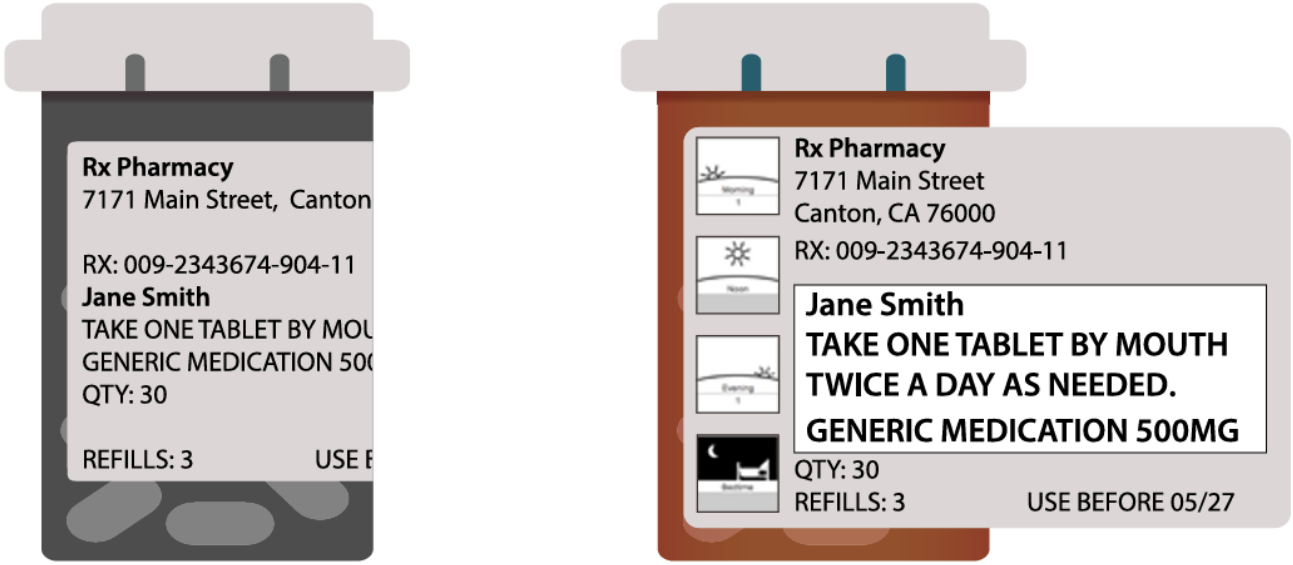
Inclusive Design

Seek out	Proactively seek out points of exclusion
Identify	Identify situational challenges (exclusion can be situational)
Recognize	Recognize personal biases (design with empathy and avoid assumptions)
Offer	Offer different ways to engage
Provide	Provide equivalent experiences (ensure everyone receives value from your design)
Extend	Extend solutions to everyone

Example

Making a Medication Schedule Easier to Use

Researchers adapted traditional written prescription labels and instructions to a universal medication schedule to make them appropriate to lower levels of literacy. As illustrated below, several elements of a patient-centered label make it more comprehensible to those with limited health literacy.



The instructions are worded more simply, there are pictorial representations of information to support comprehension, there is more white space, and the font size is larger. These evidence-based strategies have been shown to improve medication comprehension and adherence in randomized controlled trials.

Sources: William H Shrank, Ruth Parker, Terry Davis et al. 2010, "**Rationale and Design of a Randomized Trial to Evaluate an Evidence-Based Prescription Drug Label on Actual Medication Use.**" *Contemp Clin Trials* 31 (6): 564-71; and Michael S Wolf, Terry C Davis, Laura M Curtis et al. 2016. "A Patient-Centered Prescription Drug Label to Promote Appropriate Medication Use and Adherence." *J Gen Intern Med* 31 (12): 1482–1489.

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Inclusive design supports:

- **Recognition & inclusion:** When the design process accounts for the context and needs of as many people as possible
- **Power & Harmony:** When the design process includes 'affected' users in decision-making process
- **Well-being:** When the designed product promotes and protects health (think Daniels)



Pratt (reading to come in Week 4)

Part of **socially responsible science** (e.g., get diverse perspectives early, incorporate underlying social/historical contexts)

TEN SIMPLE RULES FOR SOCIALLY RESPONSIBLE SCIENCE

Zivony et al. (142) proposed the following rules for researchers aiming to conduct socially responsible science:

1. Get diverse perspectives early on.
2. Understand the limits of your design with regard to your claims.
3. Incorporate underlying social theory and historical contexts.
4. Be transparent about your hypothesis and analyses.
5. Report your results and limitations accurately and transparently.
6. Choose your terminology carefully.
7. Seek a rigorous review and editorial processes.
8. Play an active role in ensuring correct interpretations of your results.
9. Address criticism from peers and the general public with respect.
10. When all else fails, consider submitting a correction or a self-retraction.

Inclusive Design & Rawls

Just as Rawls asks us to consider designing our society to benefit its least advantaged members, inclusive design asks us to design for the least privileged members of society.



How to identify who to inclusively design for?

- **Vertical dimensions of inequality**
  - e.g., inequalities between the rich and poor

- **Horizontal dimensions of inequality**
  - e.g., inequalities between groups distinguished by culturally-defined differences such as race, gender, and religion.
  - Science can become a basis for culturally-defined difference that shapes the distribution of access to decisions.

(Harsh, 2011)



Inclusive Design & Daniels

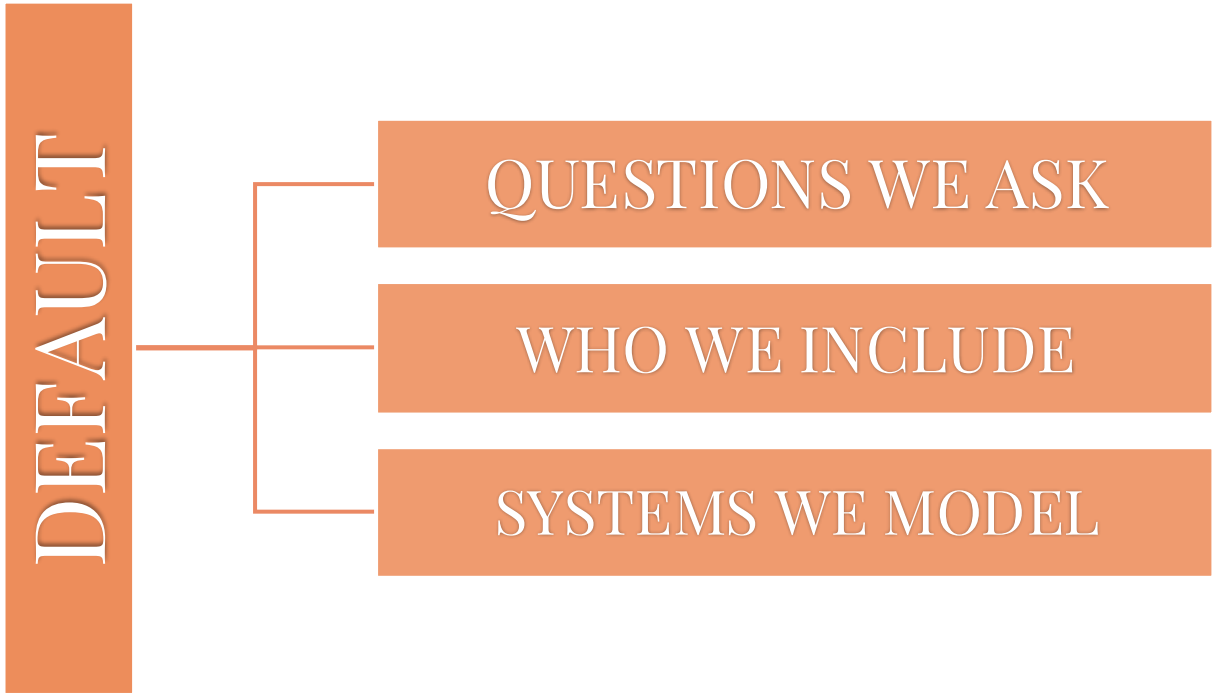
A **fair and deliberative process** can help secure health justice and participatory justice. Bringing this process into inclusive design makes for more inclusive outputs.



What we accept & reward as the default matters

We make assumptions when modeling human health or designing tech.

This is different from what we accept as the default.



Justice is key when thinking about the development of health tools and technologies for complex systems outside of the “default”.

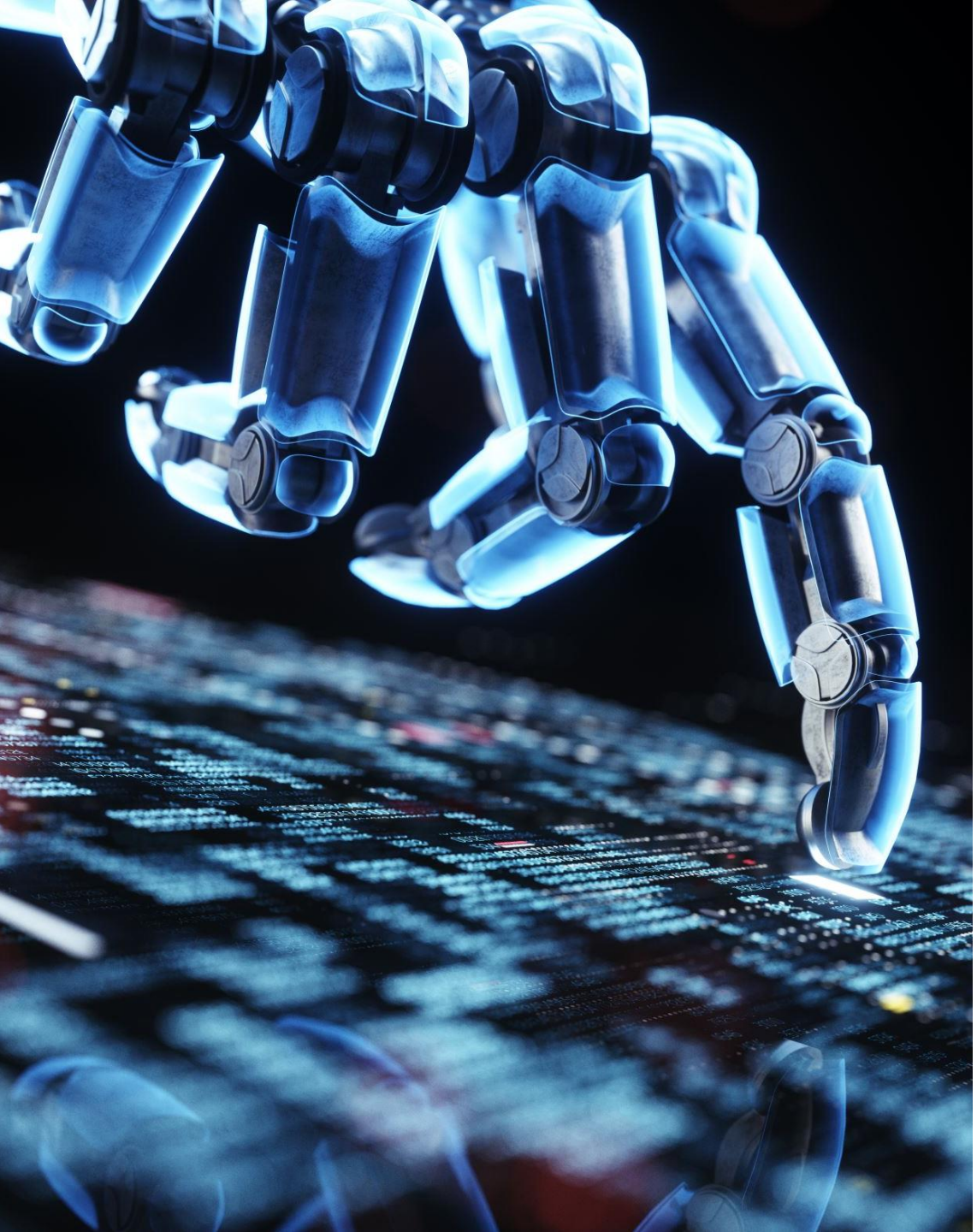
In summary:



- Technology is a designed tool that permits an action (e.g., solving or creating problems, intervening in the environment)

- Assumptions, beliefs, and values inform tech development, and other aspects of a society, in ways that can create injustice
  - "*Activities done in the name of participation can maintain or reinforce social hierarchies*" (Harsh, 2011)
- Inclusive design benefits ***everyone***, not just those who have been thus far excluded

Questions?



Looking to next Monday:

How are science & technology used to create difference? What does this mean for justice?

If extra time

You are a tech developer trying to develop “MediReach”: a telehealth app that connects patients, community health workers, and remote doctors. What are the main issues, steps, and choices you would consider in designing this product? How you will ‘access’ your intended communities?

Now imagine that you are a member of the community that has been approached by the developers. Discuss the main issues, steps, and choices you would consider in designing this product.

Notice any differences between what you considered as the researcher and what you considered as the community member?